

Smart Library Loan Ontology

Final presentation for a provenance-aware knowledge graph supporting circulation, validation, and future natural-language access.

**From library records to a
queryable semantic system**

Problem Definition and Motivation

Library knowledge is fragmented across catalog records, digital resources, and operational transactions.



Real-world problem

Traditional library data are spread across metadata records, circulation logs, spreadsheets, and digital platforms, making integrated querying difficult.



Why ontology

An ontology provides shared vocabulary, explicit semantics, and reusable relations for resources, people, transactions, topics, and provenance.



Project goal

Build a smart-library knowledge graph that supports competency questions, SHACL validation, and a future LLM interface grounded in graph results.

Project Scope and Final Deliverables

The final submission combines ontology modeling, data acquisition, validation, documentation, and reproducibility artifacts.

25

Ontology classes

Phase 2 ontology count

23

Object properties

Including provenance and profile links

16

Datatype properties

Including confidence and interaction fields

System Architecture

The project follows a staged semantic workflow inspired by ontology-centered automation research.



Specification

Competency questions define what the ontology and graph must answer, including circulation, provenance, and recommendation-ready retrieval.



Acquisition + graph

Data are collected from APIs and seed tables, normalized into semantic topics and metadata records, and represented as RDF triples.



Access + control

SPARQL queries retrieve evidence, SHACL validates quality, and a future LLM layer can translate natural-language questions into graph-grounded answers.

Ontology Design and Modeling Decisions

The design separates a stable circulation core from newly added provenance and personalization modules.



TBox core

Core classes model members, librarians, resources, branches, publishers, loans, reservations, and resource statuses.



Phase 2 extension

New classes such as DataSource, MetadataRecord, Topic, MemberProfile, InteractionEvent, and Recommendation extend the ontology without breaking the original model.



ABox examples

Individuals such as Book_OntologyDesign, MetadataRecord_B5001, Profile_Aylin_Demir, and Recommendation_001 demonstrate real instance-level use.

Ontology Expansion and Reuse

Version 0.2.0 moves beyond a simple draft by adding reusable semantic modules and lightweight vocabulary alignment.

476

RDF triples

Current graph size

25

Named individuals

ABox instances in the current graph

v0.2.0

Ontology version

Phase 2 extension

Data Acquisition and Mapping

Phase 2 makes the data workflow explicit instead of treating instances as manually inserted examples only.



Data sources

Open Library API and Google Books API provide bibliographic enrichment, while curated seed tables stabilize identifiers, status labels, and topics.



Preprocessing

The workflow normalizes identifiers, resource states, topic labels, and metadata-source references before RDF population.



Ontology mapping

Rows are mapped to LibraryResource, MetadataRecord, DataSource, MemberProfile, and InteractionEvent individuals, making provenance queryable.

Knowledge Graph Construction

The ontology and the populated data together form a queryable RDF graph rather than an isolated OWL schema.



Representative triples

Loan_001 loanedTo
Member_Aylin_Demir,
Book_OntologyDesign hasStatus
OnLoan, MetadataRecord_B5001
retrievedFromSource Open Library
API.



Graph modules

The final graph integrates
circulation, bibliographic
metadata, provenance, semantic
topics, and recommendation-ready
profile signals.



Deployment logic

The graph can be loaded into
Protégé or a triplestore such as
GraphDB, queried with SPARQL, and
validated with SHACL.

SPARQL Queries and Competency Questions

The query layer covers retrieval, provenance, personalization, and aggregation scenarios.

12

SPARQL queries

Across three query types

4

Topic-resource links

Returned by resource-topic query

0.93

Recommendation score

For Aylin Demir

Validation and Data Quality

SHACL makes the expected integrity rules explicit and testable.



Shapes defined

LoanShape, ReservationShape, LibraryResourceShape, MetadataRecordShape, and RecommendationShape were implemented for core integrity checks.



Constraint logic

The validation layer enforces cardinality, class membership, mandatory properties, and confidence-score bounds.



Result

The current validation report conforms successfully, showing that the final graph satisfies the SHACL constraints defined for the project.

LLM Integration Workflow

The project prepares a graph-grounded natural-language access layer rather than an unverified free-text answer system.



Input to SPARQL

A user asks a natural-language question, and the system maps it to a SPARQL template using ontology vocabulary and competency-question patterns.



Graph grounding

The knowledge graph returns structured results, which become the evidence layer for any generated answer.



Hallucination mitigation

The answer should be restricted to retrieved graph facts; if no evidence exists, the system should explicitly state that no matching graph result was found.

Evaluation, Limitations, and Final Deliverables

The final project is academically complete, reproducible, and ready for presentation and submission.

2

Final PDFs

Report + specification

1

Public repository

With ontology, queries, SHACL, docs

1

Widoco site

Hosted documentation target

Repository and Documentation Access

The full submission package is publicly accessible and organized for reproducibility.



GitHub repository

github.com/kemkem042/SmartLibraryOntology contains the ontology files, queries, SHACL shapes, report artifacts, presentation, results, and scripts.



Widoco documentation

kemkem042.github.io/SmartLibraryOntology/widoco/ provides ontology metadata, class hierarchy, properties, namespaces, and diagrams.



Submission package

The repository also includes report PDF, specification PDF, presentation PDF or PPTX, query results, and SHACL validation outputs.